

ORIGINAL INVESTIGATION

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Computed tomographic features of confirmed gallbladder pathology in 34 dogs

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EQUATOR network disclosure: No EQUATOR checklist was used.

Abstract

Gallbladder pathology is common in dogs, but published studies describing the computed tomographic (CT) appearance of many gallbladder pathologies are currently lacking. This retrospective, multicenter, cases series, descriptive study evaluated the CT features of confirmed gallbladder pathology in 34 dogs. In this subset of dogs, the most common pathologies included cystic mucosal hyperplasia (15/34, 44.1%), gallbladder wall edema (9/34, 26.5%), gallbladder mucocele (8/34, 23.5%), bactibilia (7/34, 20.6%), cholecystitis (6/34, 17.6%), white bile (6/34, 17.6%), and cholelithiasis (4/34, 11.8%). The presence of intraluminal nodules, gallbladder wall thickening, hyperattenuating material (35–100 HU), and mineral attenuating material (>100 HU) were the most common abnormalities detected. However, overlap of each of these findings with a variety of gallbladder pathologies showed that none of the findings were pathognomonic for any of these pathologies. The presence of any of these CT abnormalities should increase the suspicion of gallbladder pathology and prompt further evaluation of the gallbladder for definitive diagnosis.

KEYWORDS

bile culture, biliary, canine, cytology, histopathology

1 | INTRODUCTION

More veterinary patients are being imaged with computed tomography (CT) due to increased accessibility and improved detection of pathology, especially in large breed dogs.¹ In regard to the gallbladder in particular, extensive research evaluating the normal volume,^{2,3} imaging appearance,⁴ and appearance of specific pathologies^{5–11} has been performed using ultrasound. Similar studies using CT are limited^{12–15} and while attempts may be made to extrapolate from the data of

ultrasound research, the direct correlation between the two modalities needs to be validated or research focused on CT is needed. For example, the recently published study evaluating gallbladder mucoceles showed that the CT appearance lacks the same classic stellate or striated appearance described in ultrasound.^{9,12}

Computed tomography evaluation of the gallbladder and common bile duct (CBD) was performed in both normal Beagles and dogs with diseases that may cause cholestasis.¹³ It was noted that the gallbladder volume and CBD diameter as assessed on ultrasound were not significantly different from CT.¹³ In regard to the use of CT for evaluation of gallbladder mucoceles, dogs with mucoceles have higher attenuating gallbladder luminal contents and often have centrally distributed mineral as compared to dogs with gallbladder sludge or no gallbladder sludge.¹² For evaluation of the biliary tract using CT cholangiography,

Abbreviations: CT, computed tomography; CBD, common bile duct; CMH, cystic mucosal hyperplasia; ROI, region of interest.

Previous presentation or publication disclosure: Portions of this study were presented at the Purdue Resident Seminar Series October 25, 2019 and as a poster for the Purdue Veterinary Medicine Research Day April 2020.

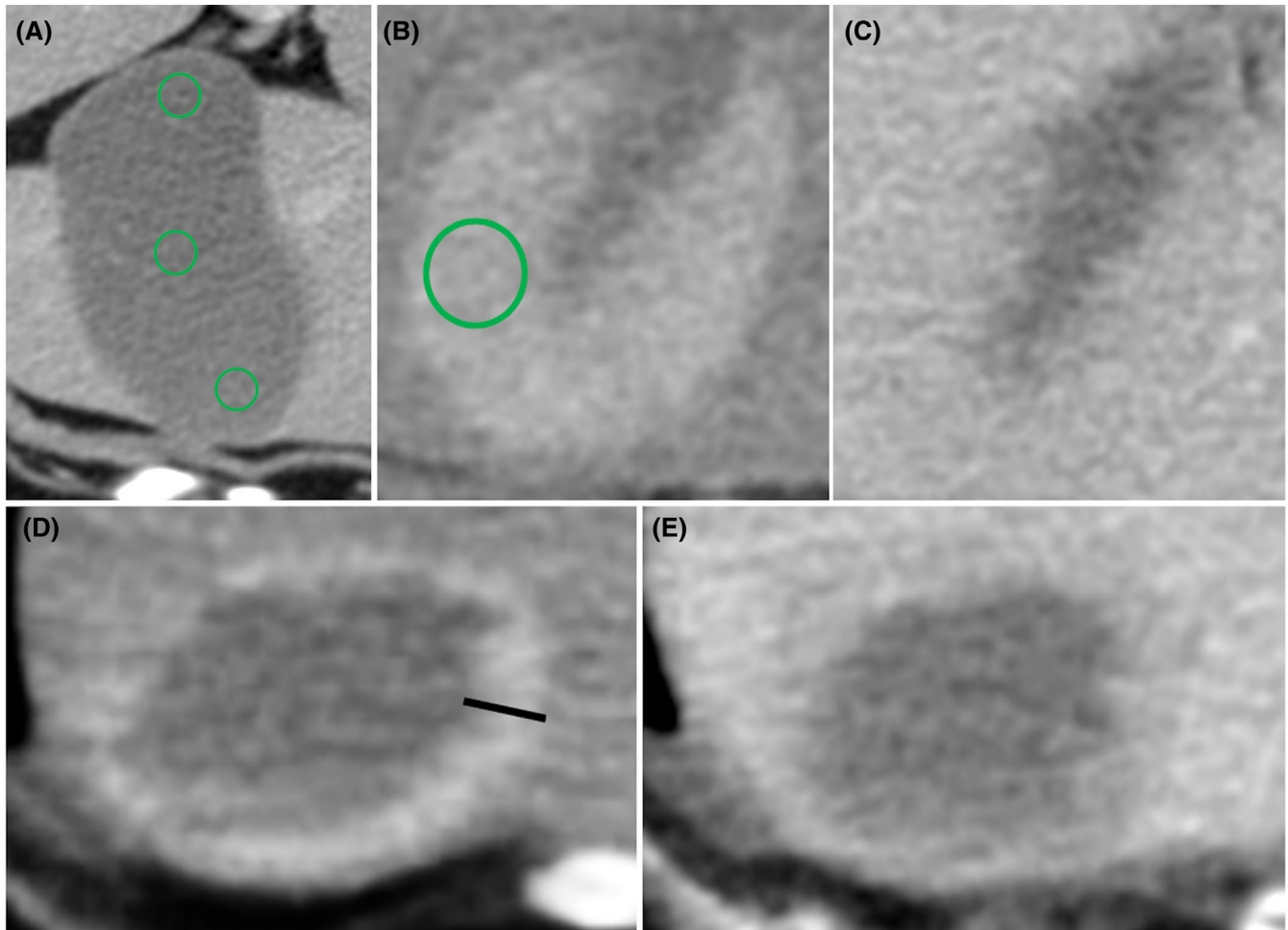


FIGURE 1 Transverse images, precontrast (B, D) and postcontrast (A, C, E), in soft tissue window (window level: 40, window width: 350); slice thickness: (A) 2.5 mm, (B-E) 5 mm; algorithm: (A) detail, (B-E) soft tissue. A, Normal gallbladder wall thickness in a dog with white bile. The wall is too thin to be accurately measured. Green circles illustrate the way the three regions of interest (ROI) were placed for measurement of the HU of the gallbladder contents. B and C, Hyperattenuating material causes the impression of either gallbladder wall thickening or a large volume of adhered material in this dog with a gallbladder mucocele, CMH, bactibilia, and cholecystitis. Green circle illustrates the way a ROI of the hyperattenuating material was performed. D and E, Thickened gallbladder wall with black line highlighting the maximum wall thickness in a dog with a gallbladder mucocele, CMH, and gallbladder wall edema. The attenuation of the wall is 88HU. In this dog, the reviewers selected that gallbladder wall visualization was best pre-contrast and used the pre-contrast image to measure wall thickness [Color figure can be viewed at wileyonlinelibrary.com]

gadoxetic acid was reported to have good enhancement of the biliary tract in healthy dogs, and a separate study reported good visualization of the biliary tract and biliary contrast excretion into the duodenum using meglumine iotroxate in dogs with gallbladder mucocoeles without complete biliary obstruction.^{14,15} This shows the use of contrast can aid in delineation of the biliary tract and gallbladder,^{14,15} but aside from the two referenced studies for gallbladder mucocoeles¹² and dogs with diseases that may cause cholestasis,¹³ references are lacking for the appearance of gallbladder pathology on CT with and without intravenous contrast administration. Given the limited research on the CT appearance of gallbladder pathology, the purpose of this study was to describe the CT appearance of confirmed gallbladder pathology in dogs.

2 | MATERIALS AND METHODS

2.1 | Study design and inclusion criteria

This was a retrospective, multicenter, case series, descriptive study. The data were acquired from dogs treated using standard veterinary practices. As a retrospective study, approval for use of patient data was granted by the hospital directors and an animal care and use protocol was not required. Medical records from January 1, 2011 to November 1, 2018 were searched using the databases of the Purdue University Veterinary Teaching Hospital and the University of Florida Small Animal Hospital to identify all dogs with a final diagnosis including one or more of the following words: gallbladder, gall bladder, cholecystitis,

TABLE 1 Computed tomographic features of gallbladder pathologies

Pathology	n	Nodule or mass	Wall thickening	Hyper-attenuating material	Mineral attenuating material	Pneumo-bilia	Defect in wall	Effusion	None
Bactibilia	7	3 (42.9%)	1 (14.3%)	4 (57.1%)	2 (29.3%)	-	-	4 (57.1%)	-
Carcinoid	1	1 (100%)	-	1 (100%)	-	-	-	-	-
Cholecystitis	6	2 (33.3%)	1 (16.7%)	4 (66.7%)	3 (50%)	1 (16.7%)	-	1 (16.7%)	-
Cholelithiasis	4	-	2 (50%)	3 (75%)	4 (100%)	-	-	-	-
CMH	15	4 (26.7%)	9 (60%)	8 (53.3%)	10 (66.7%)	-	-	3 (20%)	2 (13.3%)
Gallbladder sludge	2	1 (50%)	1 (50%)	-	2 (100%)	-	-	-	-
Hematoma	1	-	1 (100%)	1 (100%)	-	-	1 (100%)	-	-
Infarction	1	-	1 (100%)	1 (100%)	1 (100%)	-	-	-	-
Intraluminal plant material	1	-	1 (100%)	-	1 (100%)	1 (100%)	-	-	-
Leiomyoma	1	-	-	1 (100%)	-	-	-	-	-
Mineralization of the wall	1	-	1 (100%)	1 (100%)	1 (100%)	-	-	-	-
Mucocele	8	3 (6.3%)	5 (62.5%)	5 (62.5%)	6 (75%)	-	-	1 (12.5%)	-
Wall edema	9	1 (11.1%)	4 (44.4%)	4 (44.4%)	4 (44.4%)	-	-	1 (11.1%)	1 (11.1%)
Wall vasculitis	1	-	-	1 (100%)	-	-	-	-	-
White bile	6	1 (16.7%)	1 (16.7%)	2 (33.3%)	4 (66.7%)	-	-	1 (16.7%)	-

Abbreviation: CMH, cystic mucosal hyperplasia.

TABLE 2 Computed tomographic features of seven nodules

Feature	Subcategories n (%)			
Enhancement	Mild 4 (57.1%)	Moderate 1 (14.3%)	Strong 2 (28.6%)	Heterogeneous 7 (100%)
Enhancement HU	Precontrast 50.67 ± 14.2 HU		Postcontrast 71.2 ± 31.5 HU	
Location	Body 3 (42.8%)	Neck 2 (28.6%)	Body and neck 2 (28.6%)	
Margination	Irregular 7 (100%)	Smooth 0 (0%)		

cholangitis, cholangiohepatitis, cholelith, cholecystolith, gall stone, choledocholithiasis, mucocele, sludge, bile duct obstruction, biliary obstruction, common bile duct, and cholestasis. Dogs were included in the study if the CT performed included the entire gallbladder and CBD to the level of the major duodenal papilla, and if confirmatory testing was performed by one of the following methods within 1 month of imaging: cytology, bile culture, gross surgery findings, gross necropsy findings, or histopathology. Exclusion criteria were time between CT and confirmatory testing greater than 1 month, normal bile verified by cytology and/or culture, normal histopathology of the gallbladder, or no comment about gross gallbladder appearance (surgery or necropsy findings).

2.2 | Clinical data

For dogs meeting inclusion criteria from either institution, the following medical record data were recorded: age at time of imaging, breed, sex, weight at time of imaging, date of imaging, date of confirmatory diagnosis, type of confirmatory diagnosis (cytology, culture, surgical findings, gross necropsy findings, or histopathology), and final diagnosis. Image acquisition data for each institution was also recorded. If a dog had more than one CT examination, only the study performed closest to confirmatory testing was included.

2.3 | Image evaluation

All of the CT examinations were retrospectively reviewed using a DICOM-viewing software (OsiriX version 5.9, 64-bit, Bernex, Switzerland). Two European College of Veterinary Diagnostic Imaging (ECVDI) board-certified radiologists (C.K.L. and H.G.H.) reviewed the images together and evaluated all studies for a consensus of imaging findings over a period of 2 weeks. Both radiologists were blinded to the final diagnosis but were aware that all included dogs had confirmed gallbladder pathology. All studies were evaluated in transverse plane using a soft tissue algorithm and a soft tissue window (window level 40, window width 350–450).

Abnormalities evaluated were limited to the gallbladder and CBD. Each case was evaluated for the presence of a nodule or mass in the gallbladder (a nodule was defined as <3 cm and a mass as ≥3 cm), gallbladder wall thickness, pneumobilia, hyperattenuating material to bile (HU 35–100), mineral attenuating material (HU > 100),¹² or a defect in the wall. If no abnormalities were identified, this was also noted.

After each individual abnormality was determined, additional features were evaluated. For nodules or masses, reviewers recorded

the location within the gallbladder (neck or body as well as intraluminal, intramural, serosal, or could not be determined), margination (smooth, irregular, lobulated, or too small to evaluate), HU, and contrast enhancement. For contrast enhancement, the HU change in attenuation (from pre- to postcontrast [highest value was used if multiple postcontrast series were present]) was classified as none, mild (increase in attenuation from 1 to 25 HU), moderate (increase in attenuation from >25 to 75 HU), or strong (increase in attenuation > 75 HU) in addition to heterogeneous, homogeneous, or peripheral. For wall thickness, findings recorded included thickness of the wall measured in millimeters (if thick enough to be measured) and if it was better visualized precontrast versus postcontrast (Figure 1). The wall was measured on either the precontrast or postcontrast images depending on which allowed for best definition of the wall margins (Figure 1). The wall was considered thickened if it was ≥1.3 mm.¹³ For hyperattenuating and mineral attenuating structures, the following details were assessed: location within the gallbladder (gravity dependent, suspended, or adhered to or within the gallbladder wall) and HU of the structure (pre- and postcontrast values were recorded for hyperattenuating material, and precontrast values recorded for mineral attenuating material). When HU values were recorded, a circular or oval region of interest (ROI) was used if the lesion was large enough, otherwise a single point was used (Figure 1). Free text descriptions were used if the radiologists' description of the lesion did not conform to the preselected choices.

The mean HU of the gallbladder contents were recorded on precontrast and postcontrast images. A single transverse image was selected that maximized the gallbladder size and, using a preset round, 10 mm² ROI, three ROIs were placed in the gallbladder: one in the most dorsal aspect, one in the most ventral aspect, and one in between (Figure 1). Reviewers were instructed to avoid intraluminal gallbladder lesions and if this could not be performed, no value was recorded.

The CBD was evaluated, and visibility was reported as not visible, part of the CBD can be traced from gallbladder to duodenum or entire CBD can be traced from gallbladder to duodenum. The maximum diameter in millimeters was recorded when visible. The presence of effusion around the gallbladder was also recorded and defined as wispy or loculated fluid attenuation.

2.4 | Statistical analysis

Data were analyzed by one of the authors (E.M.B.) after consultation with a statistician. Categorical data were reported using frequencies and percentages. The Shapiro-Wilk test was used to assess continuous variables for normality. For continuous variables that were

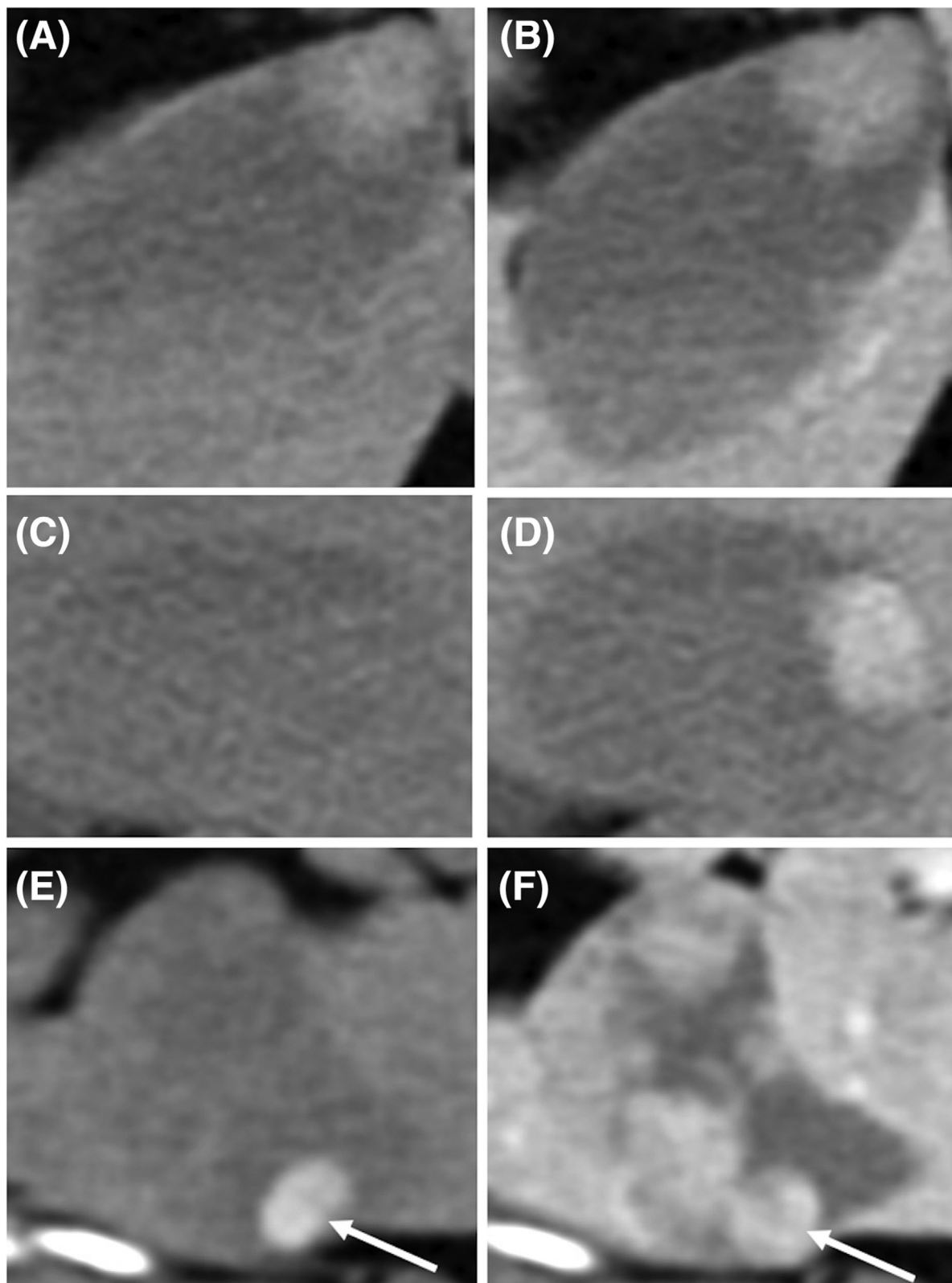


FIGURE 2 Transverse images, precontrast (A, C, E) and postcontrast (B, D, F), in soft tissue window (window level: 40, window width: 350); slice thickness: (A, B) 3 mm, (C, D) 3.75 mm, (E, F) 3 mm; algorithm: (A, B, E, F) soft tissue, (C, D) detail. A and B, An irregularly marginated nodule with mild, heterogeneous contrast enhancement located in the gallbladder neck of a dog with a gallbladder mucocele and CMH. C and D, An irregularly marginated nodule with strong, heterogeneous contrast enhancement located in the gallbladder body in a dog with a carcinoid tumor. E and F, Several irregularly marginated nodules with strong, heterogeneous contrast enhancement located in the gallbladder body as well as a mineral attenuating structure (white arrow) in a dog with cholecystitis and bactibilia

TABLE 3 Distribution of hyperattenuating and mineral attenuating contents

	N	Gravity dependent	Suspended	Adhered to or within wall	Within soft tissue nodule	HU precontrast	HU postcontrast
Hyperattenuating material	18	8 (44.4%)	9 (50 %)	4 (22.2%)	–	64.5 ± 19.6	65.5 ± 21.6
Mineral attenuating material	19	17 (89.5%)	1 (5.6%)	3 (16%)	1 (5.6%)	138, range 109–504.4	–

parametric, means and standard deviations were reported, and for those that were nonparametric, medians and ranges were reported. For dogs that had more than one postcontrast phase available, only the delayed postcontrast values were used for reporting. As a multi-institutional, retrospective study where standardization of image acquisition and contrast administration methods was not possible, statistical comparison between the reported variables was not performed. A commercial statistical software (STATA/SE 15.1; College Station, TX) was used to perform statistical analysis.

3 | RESULTS

3.1 | Study population

A total of 34 dogs met the inclusion criteria. There were 20 female spayed and 14 male neutered dogs representing 25 different breeds: 28 were pure-breed dogs and six were mixed-breed (Supporting Information 1). The mean age of the dogs was 10.5 years (± 2.7 years). The median weight was 14.3 kg (range 3.0–73.3 kg).

These 34 dogs represented 15 different gallbladder pathologies (Table 1; Supporting Information 1). The dogs with cholecystitis included necrosuppurative, lymphoplasmacytic, neutrophilic, or hemorrhagic types. Sixteen of 34 dogs (47%) had only one identified pathology: cystic mucosal hyperplasia (CMH) (4/16, 25%), white bile (4/16, 25%), choleliths (2/16, 13%), bactibilia (2/16, 13%), and one each (6%) of carcinoid, leiomyoma, and hematoma. Diagnoses were obtained via histopathology (23/34, 68%), cytology and culture (9/34, 26%), gross surgery findings (2/34, 6%), and gross necropsy findings (1/34, 3%); the total number of cases is >34 as one dog had cytology and culture as well gross necropsy findings. No dogs had cytology or culture alone. The time between imaging and sampling was a median of 1 day (range 0–20 days).

3.2 | Image acquisition parameters

All CT examinations performed at Purdue University were performed using a 4-slice (GE LightSpeed QXi) or a 64-slice (GE LightSpeed VCT), multidetector row, helical CT unit (GE, Milwaukee, WI). Scans were performed in helical acquisition mode with variable slice thickness (0.625–3.75 mm), pitch of 1, tube rotation time 0.8–1 s, 120 kVp, variable mA (240–340), and 512 × 512 matrix. All CT examinations at University of Florida were performed using an 8-slice (Aquilion 8) or a 160-slice (Aquilion Prime), multidetector row, helical CT unit (Toshiba Med-

ical Systems, Inc., Tustin, CA). Scans were performed in helical acquisition mode with 0.5 mm slice thickness, pitch of 0.8, tube rotation time 0.5 s, 120 kVp, 100 mA, and 512 × 512 matrix. The display field of view at both institutions was adjusted to dog size. Five dogs were positioned in dorsal recumbency and the remaining 29 dogs were in sternal recumbency. Transverse images were acquired in soft tissue, standard or detail algorithm.

All but two dogs had precontrast and postcontrast scans acquired, and these two cases had only postcontrast images. The contrast agent used at Purdue University was Ultravist (iopromide 300 mgI/mL, Bayer, Whippany, NJ) and at University of Florida was Omnipaque (iohexol, 300 mgI/mL, GE Healthcare, Waukesha, WI). Contrast media were injected intravenously (420–600 mgI/kg bodyweight) using either manual injection or a power injector (Purdue University-Mark V Pro-Vis, MedRad Inc., Warrendale, PA, or University of Florida-Stellant, Medrad Inc., Warrendale, PA). At Purdue University, for either manual injection or power injector use, scans performed between 2010 and 2015 had a variable flow rate and scans performed from 2016 and later varied flow rate to administer contrast over 20 s. At University of Florida, power injector flow rate was set to 2 mL/s and a pressure limit set to 325 psi. Postcontrast images varied in their timing between delayed venous phase only (13 dogs, acquired at 30–60 s after contrast administration), late delayed (1 dog, 6 min after contrast administration), or three-phase series (18 dogs, acquired at the following time points after contrast administration at Purdue University: arterial phase 20 s, portal venous phase 55 s, delayed phase 95 s, or using bolus-tracking at University of Florida).

3.3 | Computed tomographic evaluation of the gallbladder and CBD

The CT features of all gallbladder pathologies are described in Table 1 and individually described below.

Seven dogs were classified as having nodules and one dog was classified as having a mass. The features of the nodules are described in Table 2 and Figure 2. The mass occupied 80% of the gallbladder lumen and had mixed smooth and irregular margins with mild heterogeneous contrast enhancement. Of the dogs with strongly contrast enhancing nodules, one was diagnosed as carcinoid tumor, and the other was cholecystitis and bactibilia (Figure 2C–F); the carcinoid tumor had strong arterial contrast enhancement that washed out by the delayed phase (HU arterial 136, HU delayed 71).

The gallbladder wall was thick enough to be accurately measured in 15 dogs (Figure 1). The reviewers disagreed in one dog as to whether

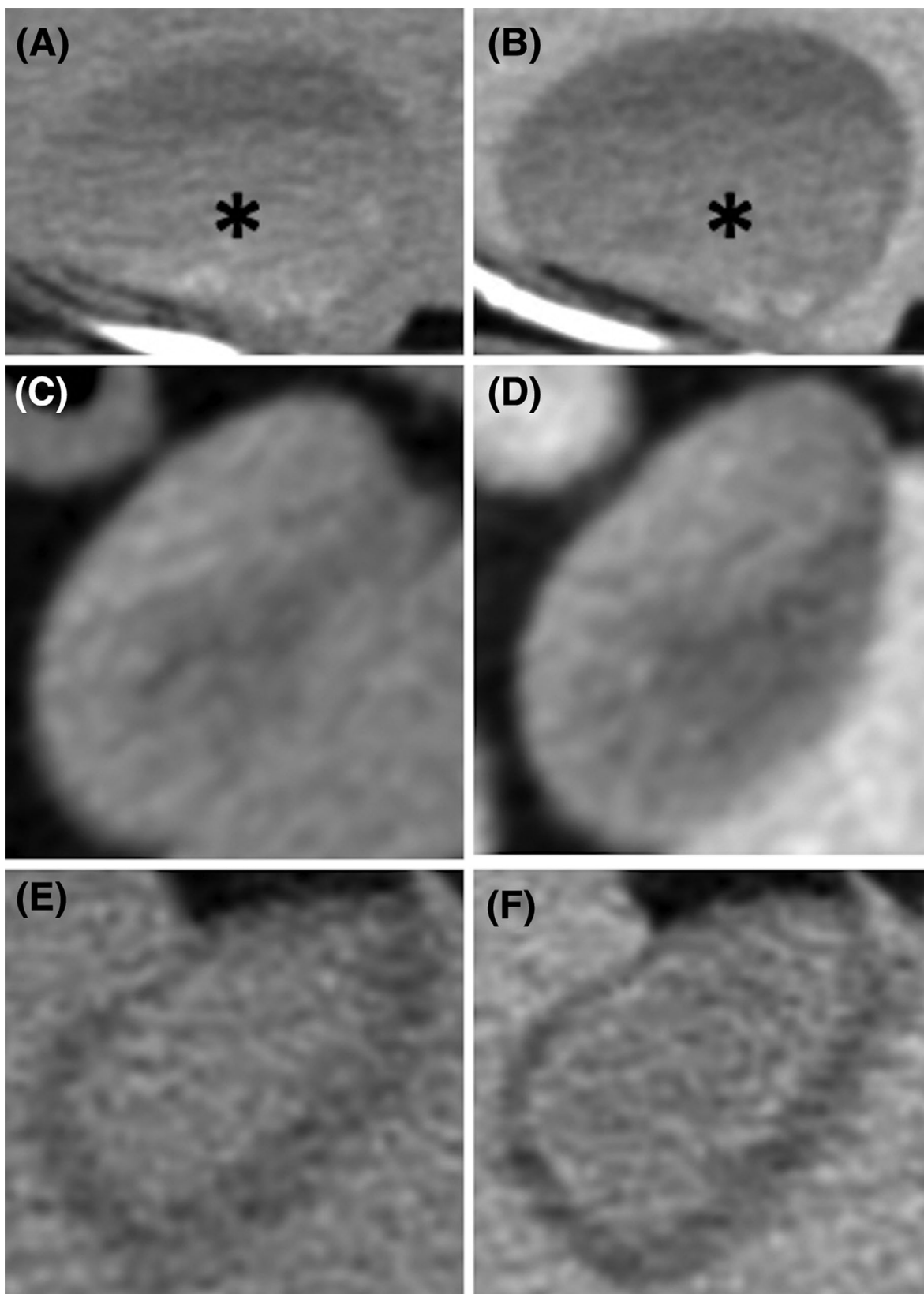


FIGURE 3 Transverse images, precontrast (A, C, E) and postcontrast (B, D, F), in soft tissue window (window level: 40, window width: 350); slice thickness: (A, B) 3 mm, (C-F) 1.25 mm; algorithm: (A-D) soft tissue, (E, F) detail. A and B, Gravity dependent location of hyperattenuating material (asterisk) in a dog with gallbladder mucocoele and CMH. C and D, Suspended hyperattenuating material in a dog with white bile. E and F, Hyperattenuating material that is suspended throughout the entire gallbladder in a dog with gallbladder wall edema and CMH

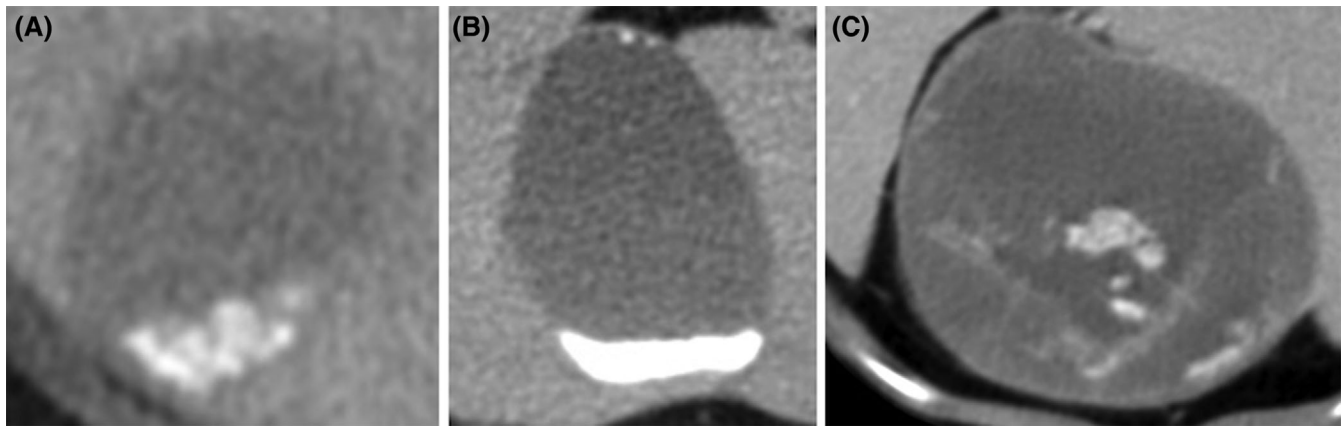


FIGURE 4 Transverse images, postcontrast in soft tissue window (window level: 40, window width: 350-450); slice thickness: (A) 0.625 mm, (B, C) 2.5 mm; algorithm: (A, B) detail, (C) standard. A, Multiple small, rounded mineral attenuating foci are present in the gravity dependent portion of the gallbladder in this dog with choleliths. B, Few small mineral attenuating foci are adhered to the dorsal aspect of the gallbladder as well as in a gravity dependent location conforming to the shape of the gallbladder in this dog with white bile and gallbladder sludge. C, Wispy mineral attenuating striations are suspended throughout the gallbladder in addition to gravity dependent locations in this dog with infarction of the gallbladder wall and choleliths

there was gallbladder wall thickening or intraluminal gallbladder material; this case was excluded from evaluation of the wall thickness (Figure 1B,C). The mean wall thickness was 3.94 mm (± 1.89 mm). The gallbladder wall was best visualized on postcontrast images in 30 of the 32 dogs (94%) that had both precontrast and postcontrast scans available; the wall of the other two dogs was best seen precontrast (Figure 1B,D).

Hyperattenuating and mineral attenuating material was detected in the gallbladder of 18 of 34 and 19 of 34 dogs, respectively (Table 3). The majority of dogs had hyperattenuating material in a suspended or gravity dependent location (Figure 3A-F) whereas dogs with mineral attenuating material more commonly had a gravity dependent location (Table 3 and Figure 4). In the dogs with material either adhered to or within the gallbladder wall, an intraluminal versus intramural location could not be differentiated. The shape of the mineral attenuating material varied from a florid-like shape to rounded to conforming to the shape of the gallbladder (Figure 4A,B). Only dog with gallbladder wall infarction had suspended mineral material (Figure 4C). The HU of the hyperattenuating and mineral attenuating material is listed in Table 3.

The HU of bile could not be recorded in any of the three locations for 10 of 34 dogs as the visible intraluminal gallbladder pathology could not be avoided. For the remaining 24 of 34 dogs, at least one value was recorded. The highest bile HU from the recorded values was selected precontrast and postcontrast, regardless of region. The median precontrast bile was 37.6 HU (range 9.2-57.3) and the median postcontrast bile was 43.6 HU (range 3.6-57.1). The mean highest precontrast and postcontrast HU of bile was reported for the most common pathologies (Figure 5).

The CBD was visible in 30 of 34 dogs, and of these, the entire duct could be traced in 21 of 30 (70%) dogs and only part of the duct was able to be traced in nine of 30 (30%) dogs. A focal area of narrowing was reported in one dog with concurrent cholecystitis and gallbladder wall edema. The maximum diameter of the common bile duct was recorded

in 29 of 34 dogs. The median maximum diameter of the common bile duct of these dogs was 4.1 mm (range 1.4-12.2 mm). Effusion surrounding gallbladder was detected in seven of 34 dogs. Pathologies seen with effusion are listed in Table 1.

4 | DISCUSSION

Fifteen different gallbladder pathologies in dogs are reported here and all have been reported previously in dogs, except for gallbladder wall hematoma (Figure 6). However, this is the first CT study to describe features and include images of many of these confirmed gallbladder pathologies. The most common gallbladder pathologies in our subset of dogs were CMH, gallbladder wall edema, gallbladder mucocele, bactibilia, cholecystitis, and white bile. Four common gallbladder CT features seen in this study were the presence of nodules, wall thickening, hyperattenuating material, and mineral attenuating material. There was a large degree of overlap of these CT features in dogs with varying gallbladder pathologies.

Of the dogs presenting with nodules or masses in the gallbladder, the most commonly represented pathologies included CMH, gallbladder mucocele, bactibilia, and cholecystitis. Cystic mucosal hyperplasia has previously been reported to have either a polypoid or sessile appearance on ultrasound⁷ and gross pathology¹⁶ so it is unsurprising that some dogs with CMH had a nodular CT appearance. As CMH is commonly seen in dogs with gallbladder mucoceles,^{7,9,17} this can also explain the presence of nodules in two of the three dogs with gallbladder mucoceles. The nodules seen in dogs with bactibilia may be due to underlying cholecystitis or another pathology that was missed as some of these dogs were diagnosed based on cytology alone. Two dogs had strongly enhancing nodules in the gallbladder and these dogs had final diagnoses of a carcinoid in one and cholecystitis with bactibilia in the other. Early wash-in and early wash-out contrast enhancement pattern

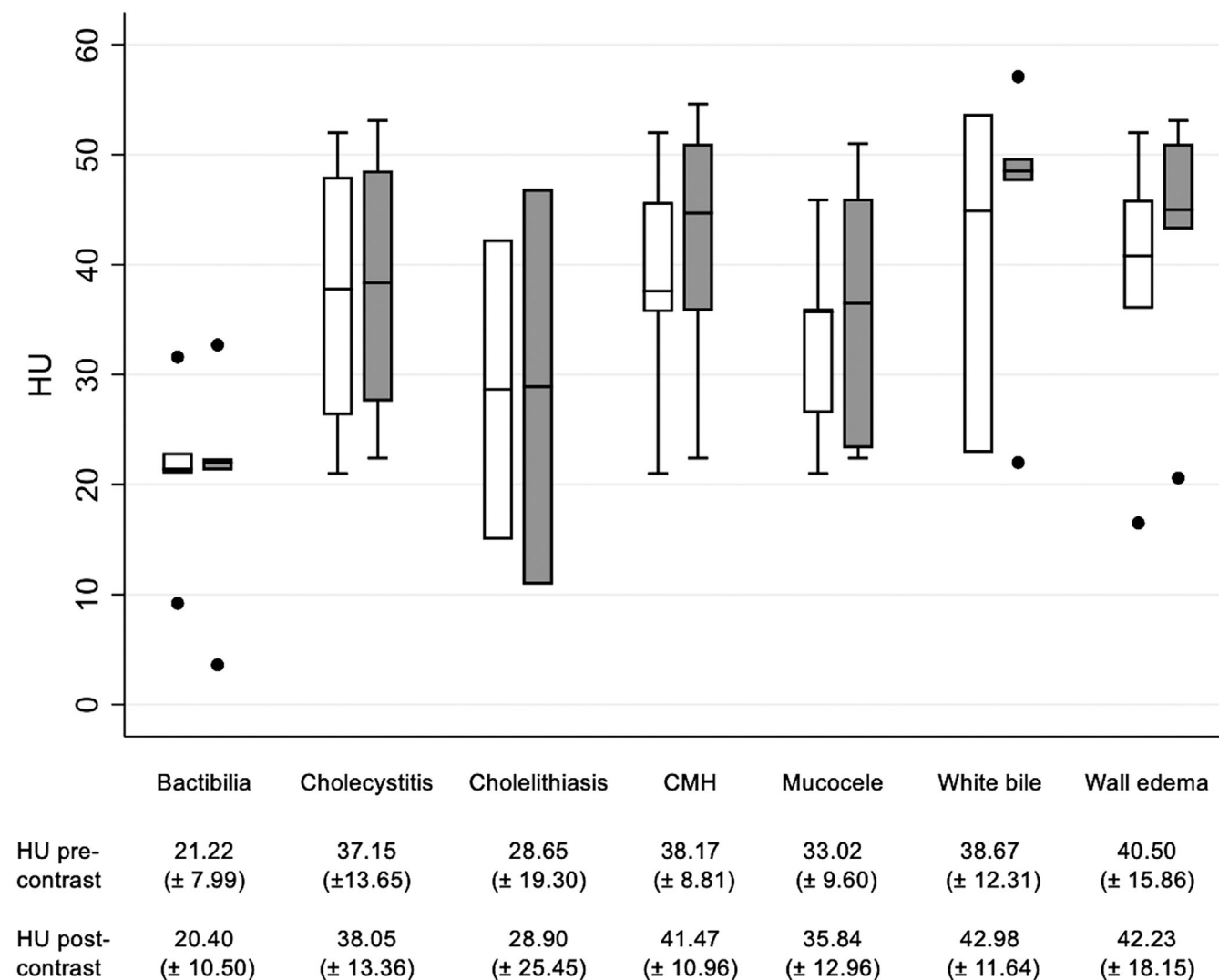


FIGURE 5 Box and whisker plot of the HU of bile precontrast (white) and postcontrast (gray) for the most common pathologies

was reported in contrast enhanced ultrasound of carcinoid tumors of the gallbladder; however, similar nodular appearance with strong contrast enhancement was not reported with cholecystitis.⁷

Gallbladder wall thickening has been described secondary to a multitude of pathologies in ultrasound⁴ and in our study was seen most commonly in dogs with CMH, gallbladder mucocoele, and gallbladder wall edema. The reported sessile distribution of CMH¹⁸ may lead to thickening of the gallbladder wall; however, the sessile form of the disease was reported to not be detectable ultrasonographically in one study.⁷ Discrimination between sessile and polypoid forms of CMH was not an objective of our study and was therefore not further subclassified. Seven of eight dogs in our study had concurrent CMH and gallbladder mucocoele, supporting the high frequency for which gallbladder mucocoele was associated with wall thickening. Gallbladder wall edema has previously been reported to cause wall thickening on ultrasound and a similar finding is supported on CT from our study.⁴

The values we report for dogs with wall thickening (3.94 ± 1.89 mm) are similar to those previously reported for dogs with diseases caus-

ing cholestasis (3.3 ± 2.76 mm).¹³ Except for one dog with white bile, the remaining dogs in our study with gallbladder wall thickening had a pathology that would be expected to contribute to increased thickness of the wall, including CMH, gallbladder wall edema, or cholecystitis. However, a total of 23 dogs had one of these three pathologies present and wall thickening was seen only in 15. This highlights that while gallbladder wall thickening supports an abnormality of the gallbladder, the absence of this finding should not rule out gallbladder pathology. An example of this is seen in Figure 7A,B in a dog with wall edema and CMH on histopathology that was classified as no CT abnormalities.

The presence of hyperattenuating material was defined as material that was 35–100 HU as this was expected to be hyperattenuating to the normal HU of bile that has been reported to be 34–35.8 HU.^{12,13} Cystic mucosal hyperplasia, gallbladder mucocoele, gallbladder wall edema, bactibilia, cholecystitis, and cholelithiasis were the pathologies most commonly associated with hyperattenuating material. As this material was seen predominantly in pathologies that cause mucin production, such as CMH and gallbladder mucocoeles, we hypothesize that this

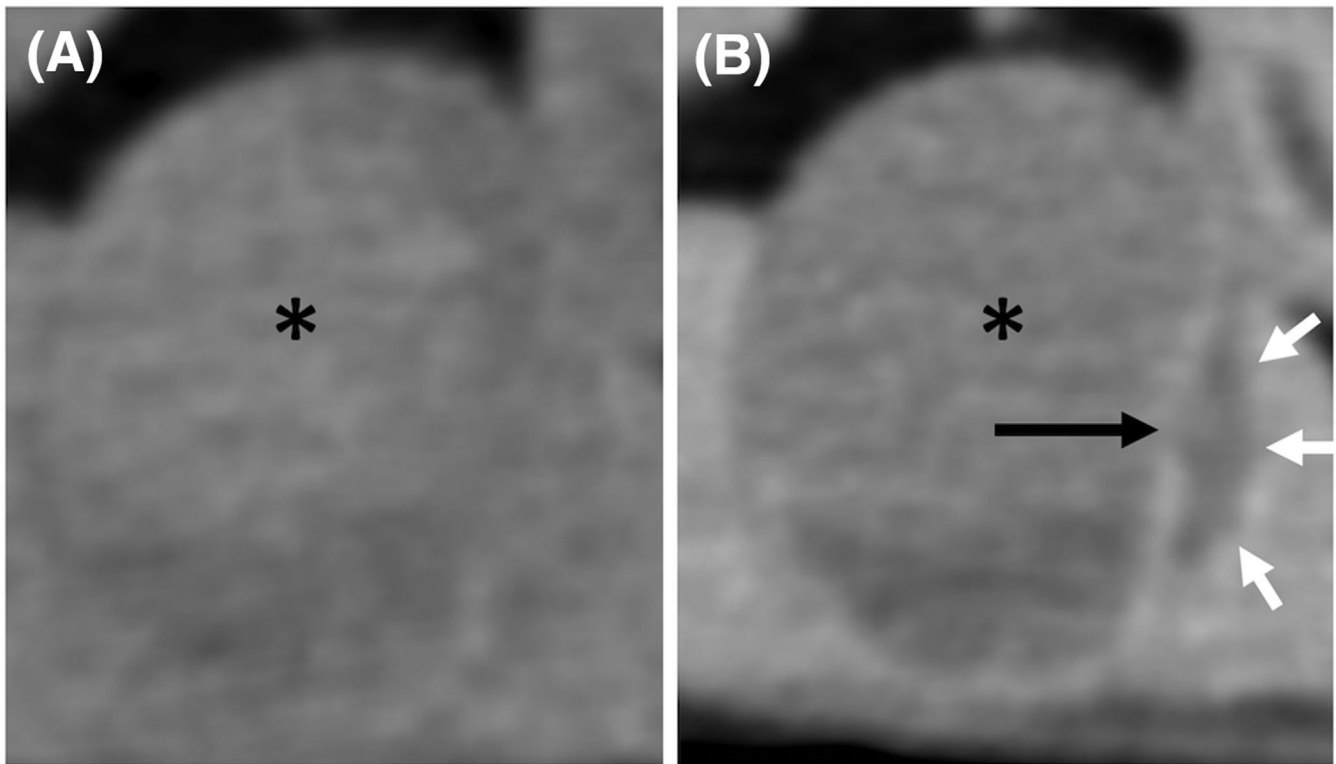


FIGURE 6 Transverse images, precontrast (A) and postcontrast (B) in soft tissue window (window level: 40, window width: 350), 3 mm slice thickness, soft tissue algorithm, of a dog with a gallbladder wall hematoma. A and B, The hyperattenuating material in the gallbladder (asterisk) was classified as suspended. B, The location of the hematoma created the impression of a focal defect in the wall (black arrow) with either focal peritoneal effusion adjacent to the gallbladder or wall thickening (white arrows)

hyperattenuating material may be due to mucinous material because of its distribution both in a gravity dependent and suspended locations throughout the gallbladder (Figure 7C,D). Similar hyperattenuating luminal contents were noted in dogs with gallbladder mucoceles.¹² Hyperattenuating material is an unexpected finding in dogs with gallbladder wall edema, bactibilia, and cholecystitis, but may be due to the presence of concurrent pathologies or gallbladder sludge that was undetected on gross pathology or cytology. In dogs with cholelithiasis, we suspect this material may have been due to poorly mineralized choleliths or concurrent sludge.

Mineral attenuating material was designated as any material >100 HU.¹² Mineral attenuation material was the most common finding of all the various pathologies evaluated. Cystic mucosal hyperplasia, gallbladder mucocele, cholelithiasis, white bile, and wall edema were the most common pathologies associated with mineral attenuating material. All of these pathologies except for gallbladder wall edema are associated with cholestasis or dysfunction of the gallbladder, which may alter the gallbladder resorption of bile salts and compounds that promote stone formation.^{16,19} The presence of this material in dogs with gallbladder wall edema is likely due to this pathology's common association with other gallbladder pathologies like cholecystitis and gallbladder mucocele. As compared to the previously reported distribution of mineral material in dogs with gallbladder mucoceles where 67% had in a central location,¹² none of our dogs with a gallbladder mucocele

had this distribution. The only dog with suspended mineral had concurrent gallbladder wall infarction and cholelithiasis. A cause for the wall infarction was unclear, which is consistent with previously reported gallbladder infarction in dogs.²⁰

The presence of abnormal intraluminal contents interfered with the radiologists' ability to measure the bile HU in multiple dogs. In observing the trends of the data, an unexpected result was the low HU of bile in dogs with bactibilia (~20 HU). Between the remainder of the pathologies present, the mean HU of bile remained between 28 and 42 HU, consistent with previous studies.^{12,13} The lower HU in dogs with bactibilia suggests a reduced density of bile. Inflammation within the gallbladder has been suggested to alter gallbladder's water absorption ability,²¹ and the presence of bacteria may lead to inflammation that could cause a less concentrated bile, lowering the HU. However, this is speculation and other unknown causes remain possible. This trend warrants further evaluation as it may be a feature to help guide use of interventions such as cholecystocentesis.

Evaluation of the CBD as well as effusion surrounding the gallbladder were performed as these areas are often assessed to help with interpretation of gallbladder pathology. The CBD was visible in the majority of the dogs and there was a wide range of diameters recorded. Previous authors have reported that the CBD of dogs as measured on CT is comparable to the measurement on ultrasound.¹³ We report a wide range of values for the CBD diameter in this subset of dogs with

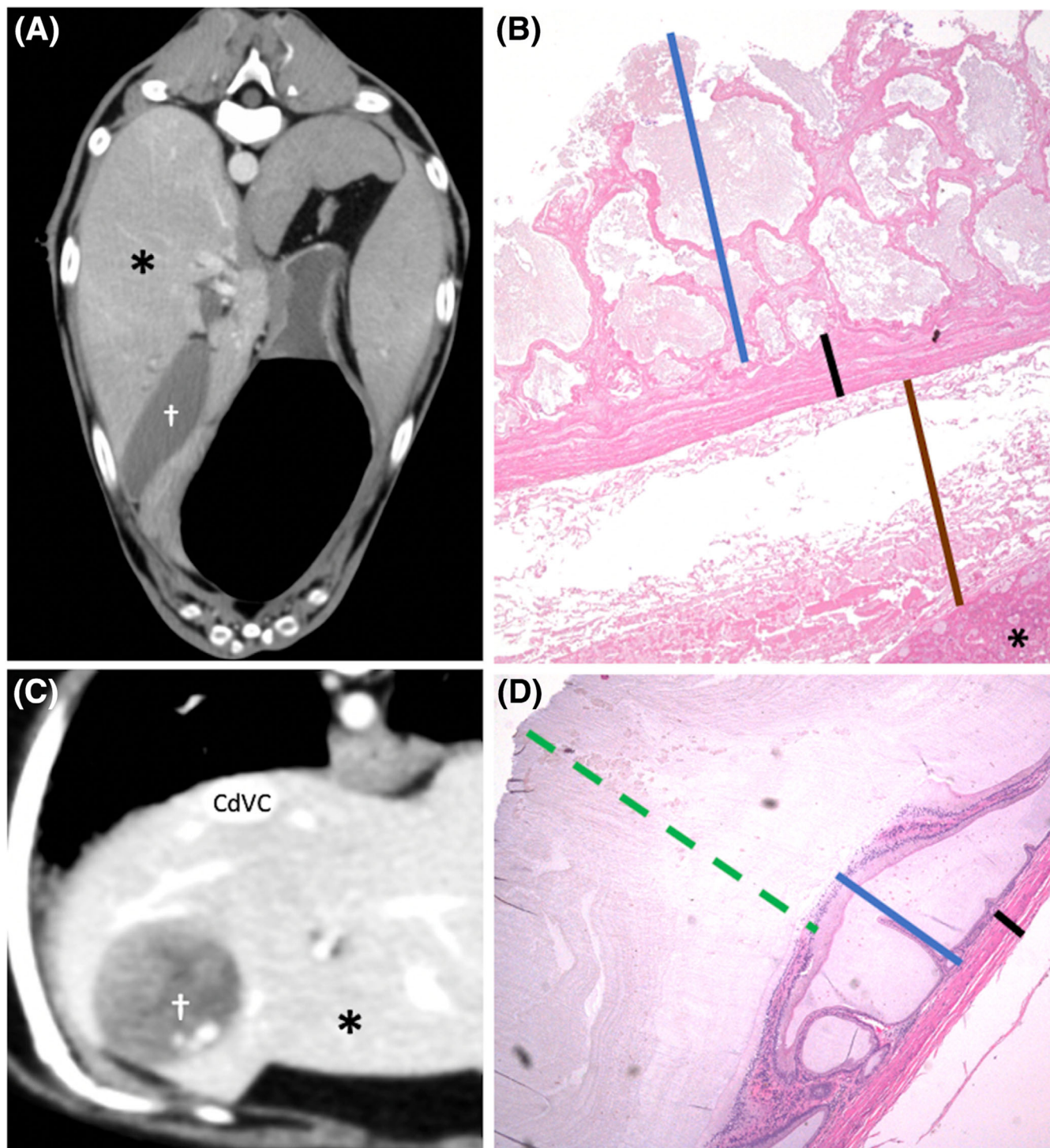


FIGURE 7 Transverse images (A, C), postcontrast in soft tissue window (window level: 40, window width: 350); slice thickness: (A) 2.5 mm, (C) 1.25 mm; algorithm (A, C) soft tissue. Histopathology images (B, D) at 4× magnification. A, This dog was classified as having no abnormalities detected on CT. B, On histopathology, there are multiple cystic cavitations expanding the mucosal layer (blue line) consistent with CMH. The muscularis layer is indicated by the black line. The serosa (brown line) is thickened secondary to gallbladder wall edema. C, This dog was classified as having hyperattenuating material and mineral attenuating material on CT. D, On histopathology, a gallbladder mucocoele was confirmed by large volumes of mucus within the lumen (dashed green line). The mucosa is expanded by mucus consistent with CMH (blue line). The muscularis and serosa are unremarkable (black line). CdCVC, caudal vena cava; dagger, gallbladder; asterisk, liver [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/vim.12909)]

gallbladder pathology. The median maximum diameter was greater than that reported for normal dogs on ultrasound and CT that may imply these pathologies also affect the CBD.^{13,22} Peritoneal effusion was present around the gallbladder in a range of pathologies.

There are several limitations of this study. To increase the total number of cases of confirmed gallbladder pathology, we included dogs with confirmation from cytology and culture, histopathology, gross necropsy, and gross surgical findings. There are limitations to the degree of pathology that can be detected using each of these methods and therefore some pathology detected by CT findings likely went undetected. This was seen in one dog where mineralization present within or adhered to the gallbladder wall on CT was not verified on histopathology. To minimize the risk of changes to gallbladder pathology between the time of imaging and confirmatory testing, we used a 1 month cutoff, but there remains the possibility that changes seen on imaging were no longer present at the time of sampling. As is present in many retrospective CT studies, there is inhomogeneity of the CT acquisition parameters for each dog. However, this is more likely to reflect the variety of CT protocols in clinical practice and it was not a goal of this paper to develop or evaluate optimal scanning protocols for the gallbladder.

We chose for our reviewers to be blinded to the type of gallbladder pathology present. This was decided as it most accurately reflected a clinical scenario where the pathology will not always be known prior to imaging. Reviewers were also blinded to the clinical patient details as the focus was on description of gallbladder pathology features and not correlation of clinical findings to gallbladder pathology. We felt that this allowed us to present a less biased view of the data.

This study serves as a reference for the types of gallbladder pathologies encountered in clinical practice as well as the variety of CT abnormalities detected with gallbladder pathology. Due to the great degree of overlap of the CT features, we did not attempt to identify any pathognomonic or defining features of any specific pathologies. The most common features of gallbladder pathology included nodules, gallbladder wall thickening, hyperattenuating material, and mineral attenuating material. The presence of any of these CT abnormalities should increase the suspicion of gallbladder pathology and further evaluation of the gallbladder using ultrasound, cholecystocentesis, repeat evaluation, or cholecystectomy may allow for definitive diagnosis.

ACKNOWLEDGMENTS

The authors acknowledge the contribution of Dr. George E. Moore for the statistical analysis of the study.

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Category 1

- (a) Conception and Design: Brand, Lim, Grosso
- (b) Acquisition of Data: Brand, Lim, Heng, Grosso, Hanlon, Jones-Hall
- (c) Analysis and Interpretation of Data: Brand, Lim, Heng, Grosso, Hanlon, Jones-Hall

Category 2

- (a) Drafting the Article: Brand
- (b) Revising Article for Intellectual Content: Brand, Lim, Heng, Grosso, Hanlon, Jones-Hall

Category 3

- (a) Final Approval of the Completed Article: Brand, Lim, Heng, Grosso, Hanlon, Jones-Hall

CONFLICT OF INTEREST

The authors declare no potential conflict of interest with respect to the research, authorship, and/or publication of this article.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Brand EM, Lim CK, Heng HG, Grosso FV, Hanlon J, Jones-Hall Y. Computed tomographic features of confirmed gallbladder pathology in 34 dogs. *Vet Radiol Ultrasound*. 2020;61:667–679.
<https://doi.org/10.1111/vru.12909>